Q No 33

Explain Chi-Square method?

The chi-square distribution contains only one parameter, called the number of degree of freedom (df), where the term d.f represents the number of independent random variable that expresses the chi-square.

$$\chi^2 = \sum_{i=1}^{n} \frac{(|o-e|-e)^2}{}$$

Q.No30 The department claims that the exceeds Rs. 2500 at the 0.05 level of, then formulate null alternative hypothesis?

Formulation:

$H_0 : \mu < 2500$

$H_1 : \mu > 2500$

Q.No31 When expected frequency in any category is less than 5 what we use?

If an expected frequency in any category is less than 5, then we use Yate' correction of continuity

$$\chi^2 = \sum_{i=1}^{2} \frac{(|O-E| - \frac{1}{2})^2}{E}$$

it is applied in

it is applied in the normal approximations to the binomial distribution.

Q.No 26 If TSS = 272.25

$\sum_i \sum_j x^2 = 4953$

Find (C.F) ~~Coefficient~~ correction factor -

we know that

$TSS = \sum_i \sum_j x_{ij}^2 - C.F$

$272.25 = 4953 - C.F$

$C.F = 4953 - 272.25$

$C.F = 4680.75$

Q.No 29 what is relation b/w mean and variance in poisson distributi

The mean and variance in the poisson distribution are same i.e. μ.

Q.No 30 The department claims that the exceeds Rs. 2500 at the 0.05 level of, then formulate null alternative hypothesis?

Formulation:

$H_0 : \mu < 2500$

$H_1 : \mu > 2500$

units within restrictions.

Q No26 Can we control type I error and type II error?

To control Type I error we can set simple a lower significance level.

While Type II error is impossible To control completely, The only possible chance To control Tye II error by increasing sample size.

Q No27 $n = 15$, $\alpha = 95\%$ Then find $t_{\alpha/2;v} = ?$

we are given that

$n = 15$, $\alpha = 95\% = 0.15$

$\alpha/2 = 0.95/2 = 0.075$

$V = n - 1 = 15 - 1 = 14$

یہاں $x_{0.75}$ اور $x_{0.25}$، Q_3 اور Q_1 لیں
normal distribution میں ایسے ہی لکھتے۔

Q No 24 Why we use pooled proportion instead of individual p?

We use the pooled proportion instead of individual proportion, because the pooled proportion estimate the standard error and individual not-

Q.No20 what is relationship b/w level of significance and critical region?

A critical region specifies a set of values of the test statistic for which null hypothesis is rejected.

The actual type I error that we get by using critical region it will be the level of significance-

Q.No21 what is g(x) and h(y)?

From the joint probability function for (X,Y), we can obtain the individual probability function of X and Y.

∴ individual probability function ka matlb hai X is alg or y ke alg probability

Such individual probability function are called marginal probability function.

The marginal p.f of X

$g(x) = P(X=x)$

The marginal p.f of Y

$h(y) = P(Y=y)$

Q.No22 Define point estimation-

The object of point estimation is to obtain a single number from the sample of unknown true value of population

Q No 19 $n = 1150$, $x = 450$, $P_0 = 0.39$

$H_0 : P_0 = 0.39$, $\alpha = 0.01$

Stated hypothesis.

1- Formalation

$H_0 : P_0 = 0.39$

$H_1 : P_0 \neq 0.39$

2- LOS

$\alpha = 0.01$, $\alpha/2 = 0.01/2 = 0.005$, $1 - \frac{\alpha}{2} = 0.995$

3- Test statistic

$$z = \frac{x - np_0}{\sqrt{np_0 q_0}}, \quad z = \frac{(x \pm 1/2) - np_0}{\sqrt{(np_0 q_0)}}$$

4- Critical region

$$z_{1-\alpha/2} = z_{0.995} = 2.576$$

$$|z|_c < 2.576$$

5- Calculation

we are given that

$n = 1150$, $x = 450$, $P_0 = 0.39$

$q_0 = 1 - P_0 = 1 - 0.39 = 0.61$

~~$z = 4.50$~~

$$z = \frac{(450 - 1/2) - (1150)(0.39)}{\sqrt{(1150)(0.39)(0.61)}}$$

$$= \frac{449.5 - 448.5}{16.540}$$

$$= \frac{1}{16.540} = 0.0604$$

$z = 0.0604$

6) Conclusion

Since $|z|_c < 2.576$

$0.0604 < 2.576$

It does not lies in rejection region so we do

Q No 15

Find value of χ^2 test statistic if $n = 2.5$, $s^2 = 3$, $\sigma_0^2 = 18$

we know that

$$\chi^2 = \frac{ns^2}{\sigma^2}$$

$$= \frac{(2.5)(3)}{18} = 0.4167$$

Q No 16 Write variance of hypergeometric distribution.

$$\text{Variance} = \frac{nk}{N}\left(1 - \frac{k}{N}\right)\left(\frac{N-n}{N-1}\right)$$

Q No 17 Find proportion of X_1 & X_2 if

$X_1 = 300$, $n_1 = 400$

$X_2 = 200$, $300 = n_2$

Proportion of X_1

$$P_1 = \frac{X_1}{n_1} = \frac{300}{400} = \frac{3}{4} = 0.75 = 75\%$$

Proportion of X_2

$$P_2 = \frac{X_2}{n_2} = \frac{200}{300} = \frac{2}{3} = 0.667 = 66.7\%$$

Q No 18 What is Joint distribution?

The distribution of Two or more random variables which are observed simultaneously when an experiment is performed, is called Joint distribution.

Q No 14

Let T_1 and T_2 are Two unbiased estimators. variance is

$V(T_1) = \frac{11\sigma^2}{9}$ and $Var(T_2) = \frac{13\sigma^2}{9}$

which estimator is more efficient and why?

~~We know~~

we are given that

$Var(T_1) = \frac{11\sigma^2}{9}$, $Var(T_2) = \frac{13\sigma^2}{9}$

we know that

$$E_f = \frac{Var(T_2)}{Var(T_1)} = \frac{13\sigma^2/9}{11\sigma^2/9}$$

$$= \frac{13\sigma^2}{9} \times \frac{9}{11\sigma^2}$$

$$= \frac{13}{11}$$ - which is greater Then 1, showing that $Var(T_1) < Var(T_2)$. Thus T_1 is more efficient than T_2

Q#013 Assume that X has a poisson distribution with variance 2
Calculate $P(X=2)$

we know that
The distribution function of poisson distribution is

$$P(X=x) = \frac{e^{-\mu}\mu^{x}}{x!}$$

We are given that
variance = 2
In poisson distribution
variance = mean = $\mu = 2$

So

(11)

$$P(X=2) = \frac{e^{-2}(2^2)}{2!}$$

$$= \frac{(0.1353)(4)}{2}$$

$$= 0.2706$$

Q No 10 If $E(X) = 10$, $E(Y) = 9$
find $E\left(\frac{5}{10}X + \frac{2}{6}Y\right)$

We are given that
$E(X) = 10$, $E(Y) = 9$
We have To find
$= E\left(\frac{5}{10}X + \frac{2}{6}Y\right)$

$= \frac{5}{10}E(X) + \frac{2}{6}E(Y)$

$= \frac{5}{10}(10) + \frac{2}{6}(9)$

$= 5 + 3 = 8$

Q No 11 In a normal distribution
How much data that fall in interval
$\mu \pm 3\sigma$

In normal distribution 99.73% dat fall in the $\mu \pm 3\sigma$ interval

یہاں پہ $\mu \pm \sigma$ اور $\mu \pm 2\sigma$ بھی لکھ رہی ہوں تو بھی آ جائے گا۔

$\mu \pm \sigma = 68.27\%$
$\mu \pm 2\sigma = 95.45\%$

Q#09 Critical region for following hypothesis

$H_0 : \sigma^2 = 25$

$H_1 : \sigma^2 \neq 25$

where $\alpha = 0.05$, $n = 25$

Critical region

$|z| < z_{1-\alpha/2}$

~~$1-\alpha$~~ $\alpha = 0.05$, $\alpha/2 = 0.025$

$1 - \frac{\alpha}{2} = 0.975$

$z_{1-\alpha/2} = 1.96$

$|z| < 1.96$

X that can assume between x=1 and x=4 has a density function given by

$$f(x) = \frac{1}{3}$$

a) Show that the area under curve is equal to 1

b) Find $P(1.5 < X < 3)$

a) we have to show that

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

So

$$\int_{1}^{4} \frac{1}{3}\,dx = 1 \quad , \quad \frac{1}{3}\int_{1}^{4} dx = 1$$

$$\frac{1}{3}[x]_1^4 = 1$$

$$\frac{1}{3}[4-1] = 1$$

$$\frac{1}{3}(3) = 1$$

$$1 = 1$$

proved

b) $P(1.5 < X < 3)$

$$P(1.5 < X < 3) = \int_{1.5}^{3} \frac{1}{3}\,dx$$

$$= \frac{1}{3}\int_{1.5}^{3} dx$$

$$= \frac{1}{3}[x]_{1.5}^{3}$$

$$= \frac{1}{3}[3-1.5] = \frac{1}{3}[1.5]$$

$$= 0.5$$

Q No 7 Write Range, parameters and formulas for binomial distributions

Range

It range is from $0, 1, 2, \ldots n$

Parameters

It have only 2 parameters

n, p

Formulas

The distribution function is

$$P(X=x) = \binom{n}{x} p^x q^{n-x}$$

mean $= np$

variance $= npq$

QNo4 Name the measures of dispersion that are not based on all values?

1- Range
2- Quartile deviation

QNo5 what is a poisson distribution?

Poisson distribution is also known as rare events distribution. Poisson distribution is limiting form of binomial distribution under conditions:

* 'n' trials are efficiently large
* 'p' very small → 0
* $0.05 \leq np \leq 20$

The probability function of poisson distribution is

$$P(X=x) = \frac{e^{-\mu}\mu^{x}}{x!}$$

Ans) Given $N=10$, $n=4$ & $K=5$
find $E(X)$, $Var(X)$?

we are given that
$N=10$, $n=4$, $K=5$

&
we have to find
$E(X) = ?$, $Var(X) = ?$
we know that, the given data is in hypergeometric distribution, So

$$E(X) = \frac{nK}{N}$$

$$= \frac{(4)(5)}{10} = \frac{20}{10} = 2$$

$$\boxed{E(X) = 2}$$

$$Var(X) = \frac{nK}{N}\left(1-\frac{K}{N}\right)\left(\frac{N-n}{N-1}\right)$$

(3)

$$= \frac{(4)(5)}{10}\left(1-\frac{5}{10}\right)\left(\frac{10-4}{10-1}\right)$$

$$= \frac{\cancel{20}}{\cancel{10}}\left(\frac{5}{10}\right)\left(\frac{6}{9}\right)$$

$$= \frac{6}{9}$$

$$\boxed{Var(X) = 0.6667}$$

Q NO 15

Test the independence by a simple approach b/w intelligence of fathers and sons.

Sons	Intelligent	Non-Intelligent	Total
Intelligent	(AB) = 300	(AB) = 200	(A) 500
Not-Intelligent	(αB) = 100	(αB) = 400	(α) 500
Total	(B) = 400	(β) = 600	n = 1000

Let 'A' denotes intelligent father and 'B' denotes intelligent sons

Then

$$(AB) = 300$$

$$\frac{(A)(B)}{n} = \frac{(500)(400)}{1000} = 200$$

$$(AB) > \frac{(A)(B)}{n}$$

$$300 > 200$$

Thus there is +ve association b/w father and sons's intelligent.

Lecture 45

Q No 14

Find Coefficient of association

$AB = 55, \alpha B = 75, A\beta = 45, \alpha B = 125$

we know that

coefficient of association

$$Q = \frac{(AB)(\alpha\beta) - (A\beta)(\alpha B)}{(AB)(\alpha\beta) + (A\beta)(\alpha B)}$$

we are given that

$AB = 55, (\alpha\beta) = 75, A\beta = 45, \alpha B = 125$

$$= \frac{(55)(75) - (45)(125)}{(55)(75) + (45)(125)}$$

$$= \frac{4125 - 5625}{4125 + 5625} = \frac{-1500}{9750}$$

$$Q = -0.1538$$

This indicate negative association b/w A and B.

Q#010

if $\bar{x}=250, n=630, p_0=0.50$

then find the z-test statistic for proportion

we are given that

$\bar{x}=250$, $n=630$, $p_0=.50$, $q_0=1-0.5=0.5$

we have to find

$$Z=\frac{\bar{x}-np_0}{\sqrt{np_0q_0}}$$

$$=\frac{250-(630)(0.5)}{\sqrt{(630)(0.50)(0.5)}}=\frac{250-315}{\sqrt{157.5}}$$

$$=\frac{250-315}{12.55}=\frac{-65}{12.55}=-5.17$$

$Z=-5.17$

Q#011

1- Formulation

$H_0: \mu_1=\mu_2 \Rightarrow \mu_1-\mu_2=0 \quad (\delta_0=0)$

$H_1: \mu_1\neq\mu_2 \Rightarrow \mu_1-\mu_2\neq 0 \quad (\delta_0\neq 0)$

$\because \mu_1-\mu_2=\delta_0$

2- Level of significance

$\alpha=5\%=0.05$

$\frac{\alpha}{2}=0.025$, $1-\frac{\alpha}{2}=0.975$

3- Test statistic

$$Z=\frac{(\bar{X}_1-\bar{X}_2)-\delta_0}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}}$$

4- Critical region

$Z_{1-\frac{\alpha}{2}}=Z_{0.975}=1.96$

$|Z|>1.96$

5- Calculation

we are given that

$n_1=100$, $\bar{X}_1=1190$, $\sigma_1=90$

$n_2=75$, $\bar{X}_2=1230$, $\sigma_2=120$

we have to calculate

$$Z=\frac{(\bar{X}_1-\bar{X}_2)-\delta_0}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}}$$

$$=\frac{(1190-1230)-0}{\sqrt{\frac{(90)^2}{100}+\frac{(120)^2}{75}}}=\frac{-40}{\sqrt{81+192}}$$

$$=\frac{-40}{16.5227}$$

$Z=-2.42$

6- Conclusion

Since

$|-2.42|>1.96$

so we reject H_0 against H_1

Q No 8

Given that X is normally distributed and given the sample values $\bar{x} = 42$, $S = 5$ and $n = 20$. Find the 98% confidence interval for σ^2.

We are given that

$\bar{x} = 42$, $S = 5$, $n = 20$

$1 - \alpha = 98\% = 0.98 \Rightarrow \alpha = 1 - 0.98 = 0.02$

$\frac{\alpha}{2} = 0.01 \Rightarrow 1 - \frac{\alpha}{2} = 0.99$

$v = n - 1 = 20 - 1 = 19$

$$\frac{nS^2}{\chi^2_{\frac{\alpha}{2},v}} < \sigma^2 < \frac{nS^2}{\chi^2_{1-\frac{\alpha}{2},v}}$$

$$\frac{(20)(5)^2}{\chi^2_{0.01;19}} < \sigma^2 < \frac{(20)(5)^2}{\chi^2_{0.99;19}}$$

$$\frac{500}{36.19} < \sigma^2 < \frac{500}{7.63}$$

$13.82 < \sigma^2 < 65.53$

The required confidence interval for σ^2.

Q.No 9

Given that. X is normally distributed and given sample values

$\Sigma(x-\bar{x})^2 = 2.576$, and n=10. Find the 95% confidence interval for σ^2.

We are given that

$\Sigma(x-\bar{x})^2 = 2.576$, $n=10$, $\nu = n-1 = 9$

$1-\alpha = 95\% = 0.95 \Rightarrow \alpha = 1-0.95 = 0.05$

$\alpha/2 = 0.05/2 = 0.025 \Rightarrow 1-\alpha/2 = 1-0.025 = 0.975$

we have to find

$$\frac{\Sigma(x-\bar{x})^2}{\chi^2_{\alpha/2,\nu}} < \sigma^2 < \frac{\Sigma(x-\bar{x})^2}{\chi^2_{1-\alpha/2,\nu}}$$

$$\frac{2.576}{\chi^2_{0.025,9}} < \sigma^2 < \frac{2.576}{\chi^2_{0.975,9}}$$

$$\frac{2.576}{19.02} < \sigma^2 < \frac{2.576}{2.70}$$

$$0.135 < \sigma^2 < 0.95$$

Lecture 38-42

QNo1

if $n=10$ & $\alpha=0.05$, find $t_{\alpha/2;v}$

we are given that

$n=10$, $\alpha=0.05$, $t_{\alpha/2;v}=?$

we know that

$V=n-1 = 10-1=9$

$\alpha=0.05 \Rightarrow \frac{\alpha}{2} = \frac{0.05}{2} = 0.025$

$t_{\alpha/2;v} = t_{0.025;9} = +2.262$

$\boxed{t_{0.025;9} = 2.262}$: From Table of t-distribution

QNo2

If $n=5$ and $\alpha=0.05$, find $t_{\alpha/2;v}$ =

we are given that

$n=5$, $\alpha=0.05$, $t_{\alpha/2;v}=?$

we know that

$V=n-1 = 5-1=4$

$\alpha=0.05 \Rightarrow \frac{\alpha}{2} = \frac{0.05}{2} = 0.025$

$t_{\alpha/2;v} = t_{0.025;4} = 2.776$

$\boxed{t_{0.025;4} = 2.776}$

QNo3

Write down critical region for the following ~~distribution~~ hypothesis.

$H_0 : \sigma^2 = 18$

$H_1 : \sigma^2 > 18$

Critical region

~~reject $Z > Z_{1-\alpha}$~~

reject H_0 if

$Z_{(c)} > Z_{1-\alpha}$

$\therefore$ calculated

Q#04 Find upper limit :

we are given that

$\bar{X} = 400$, $\sigma = 21$, $n = 29$

$1 - \alpha = 0.95 \Rightarrow \alpha = 1 - 0.95 \Rightarrow 0.05$

$\frac{\alpha}{2} = 0.025 \Rightarrow 1 - \frac{\alpha}{2} = 0.975$

Now

$$\bar{X} \pm Z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$400 \pm Z_{0.975} \frac{21}{\sqrt{29}}$$

$$400 \pm 1.960 \frac{21}{\sqrt{29}}$$

$400 + 7.643$

407.643

Q#05

Define one tailed and Two tailed test

One tailed Test.

If the critical region is located in only on tail of the sampling distribution of test statistic, the test is called one-tailed Test.

Two-tailed Test

If the critical region is located equally in both tails of sampling distribution of Test statistic, the test is called Two-tailed test.

Lecture 33-37

Q No 1

Define an unbiased estimator.

An estimator is unbiased if the mean of its sampling distribution is equal to the population parameter to be estimated.

$$E(T) = \theta$$

OR

An estimator is defined to be unbiased if the statistic is used as a estimator has its expected value equal to the true value of population parameter being estimated

$$E(\hat{\theta}) = \theta$$

Q No 2

what is meant by confidence interval?

A confindence interval estimate of the unknown parameter θ is an interval computed from a random sample of n values $x_1, x_2, \ldots, x_n$ with a statement of how confident (e.g 90%, 95% or 99%).

we are that the interval contains the unknown parameter θ.

$$P(L < \theta < U) = 1 - \alpha \text{ for } 0 < \alpha < 1$$

L: lower limit, U: upper limit

Q No3 A computer shop contain 3 goods and 2 defective moniter. Two moniters are selected by without replacement. Find the probability that:

i) Both will be good

ii) One is good and one is defective

Solution

we are given that

good monitors = 3

defective monitors = 2

Total = 5

Selected = 2

i) Both will be good

$$P(X=2) = \frac{\binom{3}{2}\binom{2}{0}}{\binom{5}{2}} = \frac{3(1)}{10}$$

$$\boxed{P(X=2) = \frac{3}{10}}$$

ii) 1 good & 1 defective

~~P(X=1 &~~

P(X=1 is good + Y=1 is defective)

$$= \frac{\binom{3}{1}\binom{2}{1}}{\binom{5}{2}} = \frac{3 \times 2}{10} = \frac{6}{10}$$

$$\boxed{P(X=1)+P(Y=1) = \frac{6}{10}}$$

QN03 A computer shop contain 3 goods and 2 defective moniter. Two monitro are selected by without replacement. Find the probabitity that:

i) Both will be good

ii) One is good and one is defective

Solution

we are given that

good monitors = 3

defective monitors = 2

Total = 5

Selected = 2

i) Both will be good

$$P(X=2) = \frac{\binom{3}{2}\binom{2}{0}}{\binom{5}{2}} = \frac{3(1)}{10}$$

$$\boxed{P(X=2) = \frac{3}{10}}$$

ii) 1 good & 1 defective

~~P(X=1 &~~

P(X = 1 is good + ~~Y~~ = 1 is defective)

$$= \frac{\binom{3}{1}\binom{2}{1}}{\binom{5}{2}} = \frac{3 \times 2}{10} = \frac{6}{10}$$

$$\boxed{P(X=1)+P(Y=1) = \frac{6}{10}}$$

Q No 2.

if $P(X=x) = \binom{6}{x}\left(\frac{1}{2}\right)^x\left(\frac{1}{2}\right)^{6-x}$

Then find $P(X=5)$.

$$P(X=x) = \binom{6}{x}\left(\frac{1}{2}\right)^x\left(\frac{1}{2}\right)^{6-x}$$

we have to find

$$P(X=5) = ?$$

$$P(X=5) = \binom{6}{5}\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^{6-5}$$

$$= (6)\left(\frac{1}{32}\right)\left(\frac{1}{2}\right)$$

$$\boxed{P(X=5) = \frac{3}{32}}$$

Let X be a Continuous r.v with p.d.f

$$f(x) = 6x(1-x) \quad , \quad 0 \le x \le 1$$
$$= 0 \quad , \quad \text{elsewhere}$$

check that $f(x)$ is a p.d.f

The function be a density function if

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{0}^{1} 6x(1-x)\,dx = 1$$

$$\int_{0}^{1} 6x - 6x^2 = 1$$

$$6\left[\frac{x^2}{2}\right]_0^1 - 6\left[\frac{x^3}{3}\right]_0^1 = 1$$

$$\frac{6}{2}[1-0] - \frac{6}{3}[1-0] = 1$$

$$3(1) - 2(1) = 1$$
$$3 - 2 = 1$$
$$1 = 1$$

Hence proved that the function $f(x)$ is a p.d.f

Let X be a Continuous r.v with p.d.f

$$f(x) = 6x(1-x) \quad , \quad 0 \le x \le 1$$
$$= 0 \quad , \quad \text{elsewhere}$$

Check that $f(x)$ is a p.d.f

The function be a density function if

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{0}^{1} 6x(1-x)\,dx = 1$$

$$\int_{0}^{1} 6x - 6x^2 = 1$$

$$6\left[\frac{x^2}{2}\right]_0^1 - 6\left[\frac{x^3}{3}\right]_0^1 = 1$$

$$\frac{6}{2}[1-0] - \frac{6}{3}[1-0] = 1$$

$$3(1) - 2(1) = 1$$
$$3 - 2 = 1$$
$$1 = 1$$

Hence proved that the function $f(x)$ is a p.d.f

Q No 2

Find $g(2)$

y \ x	2	4	$h(y)$
1	0.10	0.15	0.25
3	0.20	0.30	0.50
5	0.10	0.15	0.25
$g(x)$	0.4	0.6	1

$g(2) = ?$

$$\boxed{g(2) = 0.4}$$

STA301

Practice Questions

lecture No 23 to 27

QNo1: Find $E(X)$

x	$f(x)$	$xf(x)$
-1	0.125	-0.125
0	0.50	0
1	0.20	0.20
2	0.05	0.1
3	0.125	0.375
		0.55

We know that

$$E(X) = \Sigma x f(x)$$
$$= 0.55$$

$$\boxed{E(X) = 0.55}$$

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STA301 QUESTIONS FILE

SOLVED BY SANAULLAH KHAN

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Subject Enrolment

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Final project (MTH620)

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Current solved paper (2021) Sta301

Question:01

Find $\hat{p}_c$ by using information.

Solution:

We are given that

$\hat{p}_1 = 0.31 \quad , \quad \hat{p}_2 = 0.24$

$n_1 = 100 \quad , \quad n_2 = 100$

$\hat{p}_c = ?$

We know that

$$\hat{p}_c = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$

putting values,

$$\hat{p}_c = \frac{100(0.31) + 100(0.24)}{100 + 100}$$

$$\hat{p}_c = \frac{31 + 24}{200} = \frac{55}{200}$$

$$\hat{p}_c = 0.275$$

Question:02

If E(x)=0.25 then find E(2x)

Solution:

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Given that ,

E(x) = 0.25

We have to find ,

E(2x) = ?

E(2x) =2E(x)

E(2x) = 2(0.25)

E(2x) = 0.5

Question:03.

Given N=10,n=4 & k=5.then find E(x) and Var(x)

Solution:-

We know that the given data is in hypergeometric distribution.

So,

KHAN ACADEMY PAKISTAN SK WORK.100%

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$$E(x) = \frac{nk}{N}$$

$$E(x) = \frac{4(5)}{10} = \frac{20}{10}$$

$$\Rightarrow E(x) = 2$$

and,

$$Var(x) = \frac{nk}{N}\left(1 - \frac{k}{N}\right)\left(\frac{N-n}{N-1}\right)$$

$$Var(x) = \frac{4(5)}{10}\left(1 - \frac{5}{10}\right)\left(\frac{10-4}{10-1}\right)$$

$$Var(x) = 2\left(\frac{5}{10}\right)\left(\frac{6}{9}\right)$$

$$Var(x) = \frac{6}{9}$$

$$\Rightarrow Var(x) = 0.6667$$

Question:04.

Write the names of the measures of dispersion that are not based on all values?

Answer:

1) **Range.**
2) **Quartile deviation.**

Question:05.

What is a poison distribution?

Answer:

A poison distribution is also known as rare events distribution.

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

Poison distribution is limiting form of binomial distribution under conditions.

These conditions are:

- ❖ **'n' trials are efficiently large.**
- ❖ **'p' very small →0 (approaches to 0).**
- ❖ **$0.05 \leq np \leq 20$**

The probability function of poison distribution is

$$P(X = x) = \frac{e^{-\mu}\,\mu^{x}}{x(\text{factorial})}$$

$$x = 0, 1, 2, 3, ..., \infty$$

Question:06.

Write the range, parameters and formulas for hypergeometric distribution.

Answer:

Range:

The range of hypergeometric distribution is 0,1,2,⋯..n ≤ N.

Parameters:

The hypergeometric distribution has three Parameters.

N, k, n.

Formulas:

The distribution function is

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

$$P(X = x) = \frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}$$

$$Mean = \frac{nk}{N}$$

$$Variance = \frac{nk}{N}\left(1-\frac{k}{N}\right)\left(\frac{N-n}{N-1}\right)$$

Question:07.

Write the range, parameters and formulas for binomial distribution.

Answer:

Range:

The range of binomial distribution is 0,1,2,…..,n

Parameters:

The binomial distribution have only 2 Parameters.

n, p.

Formulas:

The distribution function is

$$P(X = x) = \binom{n}{x} p^n . q^{n-x}$$

$$Mean = n.p$$

$$Variance = n.p.q$$

Question:08

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(03066793776)

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A continuous random variable X that can assume b/t x=1 and x=4 has a density function given by

F(x)=1/3.

a) ***Show that area under curve is equal to 1.***

b) ***Find p(1.5<x<3).***

Solution:

a) ***We have to show that***

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

by putting values,

$$\int_1^4 \frac{1}{3}dx = 1 \quad , \quad \frac{1}{3}\int_1^4 dx = 1$$

$$\frac{1}{3}|x|_1^4 = 1 \quad , \quad \frac{1}{3}(4-1) = 1$$

$$\frac{1}{3}(3) = 1 \qquad \Rightarrow 1 = 1$$

proved.

b) ***We have to show that p(1.5<x<3).***

$$p(1.5 < x < 3) = \int_{1.5}^{3} \frac{1}{3}dx$$

$$p(1.5 < x < 3) = \frac{1}{3}\int_{1.5}^{3} dx$$

$$= \frac{1}{3}|x|_{1.5}^{3} \quad , \quad = \frac{1}{3}(3-1.5)$$

$$= \frac{1}{3}(1.5)$$

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

$$\Rightarrow p(1.5 < x < 3) = 0.5$$

Proved.

Question:09.

Critical region for following hypothesis,

$H_2 : \sigma^2 = 25$, $H_1 : \sigma^2 \neq 25$

where,

$\alpha = 0.05$, $n = 25$

Solution:

Critical region

$$|Z| < Z_{1-\alpha/2}$$

$$\Rightarrow \alpha / 2 = 0.025$$
$$\Rightarrow 1 - \alpha / 2 = 0.975$$

$$Z_{1-\alpha/2} = 1.96$$
$$\Rightarrow |Z| < 1.96$$

Question:10.

If E(x)=10 , E(y)=9 then find $E\left(\frac{5}{10}x + \frac{2}{6}y\right)$.

Solution:

We have to show that

$$E\left(\frac{5}{10}x + \frac{2}{6}y\right)$$

Now,

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

$$= \frac{5}{10}E(x) + \frac{2}{6}E(y)$$

by putting values,

$$= \frac{5}{10}(10) + \frac{2}{6}(9)$$

simplyfying,

$$= 5 + 3, = 8$$

$$\Rightarrow E\left(\frac{5}{10}x + \frac{2}{6}y\right) = 8$$

Question:11

In a normal distribution how much data that fall in interval $\mu \pm 3\sigma$ **.**

Solution:

In normal distribution 99.73% data fall in the $\mu \pm 3\sigma$ **interval .**

(we can also be written as $\mu \pm \sigma \; and \; \mu \pm 2\sigma$ **and** $\mu \pm \sigma = 68.27\%$, $\mu \pm 2\sigma = 95.45\%$**).**

Question:12

A random sample of size 12 is drawn from normal population with mean = $\mu_0 = 10.9$ ***and variance*** σ^2 **.**

If $\overline{\overline{S}} = 8$, $\overline{x} = 6.4$ ***.using information complete t-statistic.***

Solution:

As we know that

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

$$t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$$

$$= \frac{6.4 - 10.9}{8/\sqrt{12}}$$

$$= \frac{-4.5}{2.309}, = -1.948$$

Question:13

Assume that X has a poison distribution with 2. Calculate P(X=2).

Solution:

As we know,

The distribution function of poison distribution is

$$P(\mathrm{X} = \mathrm{x}) = \frac{e^{-\mu}\,\mu^{x}}{x(\mathrm{fact})} \ldots\ldots\ldots(1)$$

Given,

Variance =2,

In poison distribution,

Variance=mean=μ=2

putting values in (1)

$$P(\mathrm{X} = \mathrm{x}) = \frac{e^{-2}\,(2)^{2}}{2(\mathrm{fact})}$$

$$P(\mathrm{X} = \mathrm{x}) = \frac{(0.1353)\,4}{2}$$

$$P(\mathrm{X} = \mathrm{x}) = 0.2706$$

Question:14

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

Let T_1 and T_2 are two unbiased estimators, variance is

$Var(T_1)=\frac{11}{9}\sigma^2$ *and* $Var(T_2)=\frac{13}{9}\sigma^2$ ***.Which estimator is more efficient and why?***

Solution:

We know that,

$$E_f=\frac{Var(T_2)}{Var(T_1)}=\frac{13\sigma^2/9}{11\sigma^2/9}$$

$$E_f=\frac{13\sigma^2}{9}\times\frac{9}{11\sigma^2}=\frac{13}{11}$$

13/11 is greater than 1.

Which shows that

Var(T_1) < Var(T_2) thus T_2 is more efficient than T_1.

Question:15

Find value of x^2 test statistic n=2.5, S^2 =3, $\sigma^2=18$ **.**

Solution:

As we know that,

$$x^2=\frac{nS^2}{\sigma^2}$$

putting values,

$$x^2=\frac{(2.5)3}{18}$$

$$x^2=0.4167$$

Question:16

Write the variance of hypergeometric distribution.

Ans:

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

$$Variance = \frac{nk}{N}\left(1-\frac{k}{N}\right)\left(\frac{N-n}{N-1}\right)$$

Question:17

find proportion of X_1 and X_2 if X_1 =300 , n_1 = 400

X_2 = 200 , n_2 = 300

Solution:

Proportion of X_1.

$$P_1 = \frac{X_1}{n_1} = \frac{300}{400} = \frac{3}{7}$$

$$\Rightarrow P_1 = 0.75 = 75\%$$

Proportion of X_2.

$$P_2 = \frac{X_2}{n_2} = \frac{200}{300} = \frac{2}{3}$$

$$\Rightarrow P_2 = 0.667 = 66.7\%$$

Question:18.

what is joint distribution?

Solution:

The distribution of two or more random variables which are observed simultaneously when an experiment is performed is called Joint distribution.

Question:19

n = 1150 , x = 450 , P_0 = 0.39

H_0: P_0 = 0.39 , α = 0.01

Stated hypothesis.

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FOR QUIZ CONTECT
(03066793776)

Solution:

1) ***Formulation:***

$H_0 = P_0 = 0.39$

$H_1 = P_0 \neq 0.39$

2) ***LOS***

$\alpha=0.01$, $\alpha/2=0.01/2=0.005$,

$1-\alpha/2=0.995$

3) ***Test statistic***

$$Z = \frac{X - np_0}{\sqrt{np_0q_0}}$$

$$Z = \frac{(x \pm 1/2) - np_0}{\sqrt{np_0q_0}} \quad \text{.....(A)}$$

4) ***Critical region***

$$Z_{1-\alpha/2} = Z_{0.995} = 2.576$$
$$|Z|_c < 2.576$$

5) ***Calculation***

1st of all we find q_0 from the given values,

$q_0 = 1 - p_0$, $q_0 = 1-0.39$,
$q_0 = 0.61$

now Putting the values in eq.(A)

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

$$Z = \frac{(450 - 1/2) - 1150(0.39)}{\sqrt{(1150)(0.39)(0.61)}}$$

$$Z = \frac{4495 - 448.5}{16.540}$$

$$Z = \frac{1}{16.540} = 0.0604$$

$$\Rightarrow Z = 0.0604$$

6) ***Conclusion***

Since,

$|Z|_c < 2.576$

0.0604 < 2.576

It does not lies in rejection region. So we do not reject H_0 against H_1.

Question:20

What is the relationship b/w level of significance and critical region?

Ans:

A critical region specifies a set of values of the test statistic for which null hypothesis is rejected.

The actual type I error that we get by using critical region it will be the level of significance.

Question:21

What is g(x) and h(y)?

Ans:

From the joint probability function for (x , y), we can obtain the individual probability functions of X and Y.

Such individual probability functions are called marginal probability function.

The marginal probability function of X :

g(x)=p(X=x)

The marginal probability function of Y :

h(y)=p(Y=y)

Question:22

Define point estimation.

Ans:

The object of point estimation is to obtain a single number from the sample of unknown true value of population parameter.

Question:23

Write 3 properties of normal distribution.

Ans:

1) *Continuous distribution:*

   The normal probability distribution is a Continuous distribution that ranges from $-\infty$ to $+\infty$.

   $R_x = \{x: -\infty < x < +\infty\}$

2) *Total distribution:*

   The total area under the curve is unity.

   That is $P(-\infty < x < +\infty)=1$

3) *Symmetric distribution:*

   The normal probability distribution is a Symmetric distribution.

   - Mean =median=mode
   - $X_{0.75} - \mu = \mu - X_{0.25}$
   - $\mu_1 = \mu_3 = \mu_5 = ,\ldots\ldots\ldots , = 0$

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

Question:24

Why we use pooled proportion instead of individual Proportion?

Ans:

We use the pooled proportion instead of individual Proportion, because the pooled proportion estimate the standard error and individual not.

Question:25

What is difference b/w random design and random block design?

Ans:

Complete Randomized design:

A complete randomized(CR) design , which is the simplest type of the basic designs , may be defined as a design in which treatments are assigned to experimental units completely at random , that is the randomization done without any restriction.

Randomized Complete block design:

A Randomized Complete block (RCB) design, may be defined as a design in which

I) Experimental material is divided into groups or blocks.
II) Each block contains a complete set of treatments.
III) The treatments are assigned at random to the experimental units within restrictions.

Question:26

How we can control type I error and type II error?

Answer:

To control type I error we can set a simple lower significance level.

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

While control type II error is impossible to control completely, the only possible chance to control the type II error by increasing sample size.

Question: 27

n=15, α = 95% then find $t_{\alpha/2\,;\,v}$?

Solution:

Given that

n =15 , α = 95% = 0.15

α/2 = 0.95/2 = 0.025

and,

v = n-1 = 15-1=14

so,

$t_{0.025\,;\,14}$ =2.145

Question:28:

If TSS = 272.25 , $\sum_i \sum_j X^2 = 4953$. Find (C.F) correction factor.

Solution:

We know that

$$TSS = \sum_i \sum_j X_{ij}^2 - C.F$$

Putting values,

272.25 = 4953 - C.F

C.F = 4953 - 272.25

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FOR QUIZ CONTECT
(03066793776)

C.F = 4680.75

Question:29

what is relation between mean and variance in poison distribution.

Solution:

The mean and variance in the poison distribution are same i.e. μ.

Question:30

The department claim that the exceeds Rs = .2500 at the 0.05 level , then formulate null alternative hypothesis?

Solution:

Formulation

H_0 :μ < 2500

H_1 :μ > 2500

Question:31

when expected frequency in any category is less then 5 what we use?

Solution:

If an expected frequency in any category is less than 5 , then we use Yate correction of continuity.

$$x^2 = \sum_{i=1}^{2} \frac{(|0-e|-1/2)^2}{e}$$

It is applied in the normal approximation to the binomial distribution.

Question:32.

If n=400, x=275. Then find 95% confidence interval for population proportion.

Solution:

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

We know that,

$$\hat{p} \pm Z_{\alpha/2}\sqrt{\frac{\hat{p}.\hat{q}}{n}} \text{.....(1)}$$

Now,

$$\hat{p} = \frac{x}{n} = \frac{275}{400} = 0.69$$

$$\hat{q} = 1 - \hat{p} = 1 - 0.69 = 0.31$$

$$1 - \alpha = 0.95$$

$$\Rightarrow \alpha = 0.05 \ , \ \alpha / 2 = 0.025$$

putting all values in eq.(1)

$$0.69 \pm Z_{0.025}\sqrt{\frac{(0.69)(0.31)}{400}}$$

$$0.69 \pm Z_{0.025}(0.023)$$

$$0.69 \pm (1.96)(0.023)$$

$$0.69 \pm 0.04508$$

$$\Rightarrow 0.69 + 0.04508 \ , \ 0.69 - 0.04508$$

$$\Rightarrow 0.73508 \ , \ 0.645$$

$$\Rightarrow (0.645, 0.73508)$$

Question:33.

Explain chi-square method.

Answer:

The chai-square distribution contains only one parameter, called the number of degree of freedom (d. f). where the term d. f represents the number of independent random variable that expresses the chai-square.

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

$$x^2 = \sum_{i=1}^{2} \frac{(|0-e|-e)^2}{e}$$

Question:34.

Why the statistics is called science of decision making. Explain it?

Answer:

In this it is called science of decision making because it is enabling us to establish numerical benchmarks and monitor and evaluate the progress of our policy or program.

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